Zomato Restaurant Data Analysis

Week 1 Assignment - Exploratory Data Analysis of the Zomato restaurant dataset using Python (Pandas, NumPy, Matplotlib, Seaborn).

Documentation report for the assignment notebook cohort_week_1_assignment.ipynb

1. Dataset Overview

The dataset has information about restaurants listed on Zomato, like where they are located, what cuisines they serve, how much it costs, their ratings and how many people voted.

The dataset has **9552 rows** and **22 columns**. It is read from the file `test_zomato.csv`.

Main columns I worked with, with their data types as loaded

Column	Data type
Restaurant Name	object
City	object
Cuisines	object
Average Cost for two	object
Has Online delivery	object
Has Table booking	object
Price range	float64
Aggregate rating	float64
Rating text	object
Votes	float64

Besides these, the dataset stores a few more columns like the restaurant ID, address, locality, currency, whether it is currently delivering and the rating colour, which I did not need for the analysis.

2. Data Cleaning

Before doing any analysis I cleaned up the data.

I loaded the dataset with `pd.read_csv()` and checked the structure. The shape is **9552 rows and 22 columns**.

For missing values, I found small numbers of nulls in a few columns. The numbers were small, so they did not affect the analysis much.

Missing values found per column

Column	Missing values
Cuisines	10
Price range	2
Aggregate rating	2
Rating color	2
Rating text	2
Votes	2
Locality	1

Column	Missing values
Locality Verbose	1
Longitude	1
Latitude	1
Average Cost for two	1
Currency	1
Has Table booking	1
Has Online delivery	1
Is delivering now	1
Switch to order menu	1

For duplicates, I used `df.duplicated()` and found **0 fully duplicated rows**, so there was nothing to drop.

One important thing I had to fix was the **Average Cost for two** column. It was stored as text even though it contains numbers. I converted it to a numeric type using `pd.to_numeric(..., errors="coerce")` so I could do calculations on it. After the conversion its type changed from `object` to `float64`. Only two entries were not numbers, and they became `NaN`.

The two non-numeric entries in Average Cost for two before conversion

Restaurant Name	Original value
Super Loco	NaN
Robertson Quay, Singapore River, Singapore	No

3. Data Analysis

This section answers the analysis questions from the assignment, split into the two parts as given in the assignment.

Part 1 - Data Understanding and Exploration

Q. How many unique cities are present in the dataset?
Ans. There are **142 unique cities**.

Q. Which city has the highest number of restaurants?
Ans. **New Delhi**, with **5473 restaurants**.

Q. Which cuisine appears most frequently?
Ans. **North Indian** appears the most, in **936 restaurants**.

For the statistical analysis, I used `describe()` on the numeric columns. The full summary looks like this:

Summary statistics from describe()

Statistic	Country Code	Average Cost for two	Price range	Aggregate rating	Votes
count	9552.000000	9550.000000	9550.000000	9550.000000	9550.000000
mean	18.374564	1199.326387	1.804607	2.666314	156.923037
std	56.754313	16122.023220	0.905378	1.516447	430.189709

Statistic	Country Code	Average Cost for two	Price range	Aggregate rating	Votes
min	1.000000	0.000000	1.000000	0.000000	0.000000
25%	1.000000	250.000000	1.000000	2.500000	5.000000
50%	1.000000	400.000000	2.000000	3.200000	31.000000
75%	1.000000	700.000000	2.000000	3.700000	131.000000
max	216.000000	800000.000000	4.000000	4.900000	10934.000000

The mean, median and mode of the Average Cost for two column were **1199.33**, **400** and **500**. The mean is much higher than the median, which means the data is right-skewed because of a few very expensive restaurants.

Mean, median and mode of Average Cost for two

Measure	Value
Mean	1199.33
Median	400
Mode	500

Part 2 - Data Analysis

Q. What is the average restaurant rating?
Ans. The average rating is **2.67 out of 5**.

Q. Which restaurant has the highest number of votes?
Ans. **Toit** in Bangalore, with **10934 votes**.

Q. What is the average cost for two across all restaurants?
Ans. The average is **1199.33**.

Q. Compare the average ratings of restaurants that have Online Delivery and those that do not.
Ans. Restaurants with online delivery have an average rating of **3.25**, and those without have **2.47**. So delivery restaurants are rated about **0.78 higher**.

Average rating by online delivery availability

Has Online delivery	Average rating
Yes	3.248837
No	2.465192

Q. Compare the average ratings of restaurants that offer Table Booking and those that do not.
Ans. Restaurants with table booking have an average rating of **3.44**, and those without have **2.56**. So booking restaurants are rated about **0.88 higher**.

Average rating by table booking availability

Has Table booking	Average rating
Yes	3.441969
No	2.559283

Q. What are the top 10 cities with the highest number of restaurants?
Ans. New Delhi leads by a wide margin with **5473 restaurants**. The full list is in the table below.

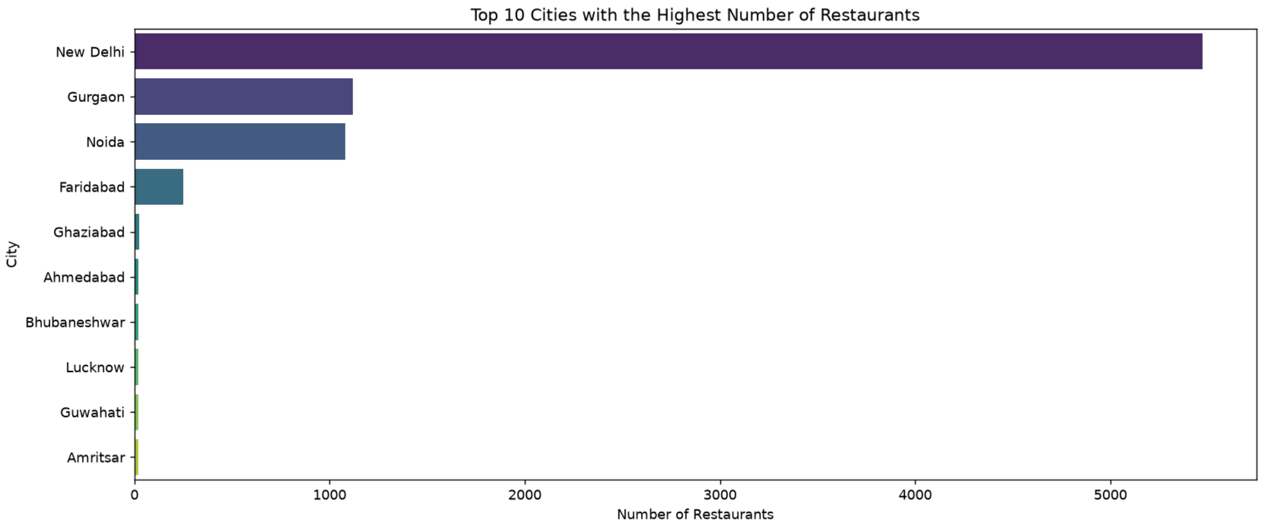
Top 10 cities with the highest number of restaurants

City	Restaurants
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Ahmedabad	21
Bhubaneshwar	21
Lucknow	21
Guwahati	21
Amritsar	21

4. Data Visualizations

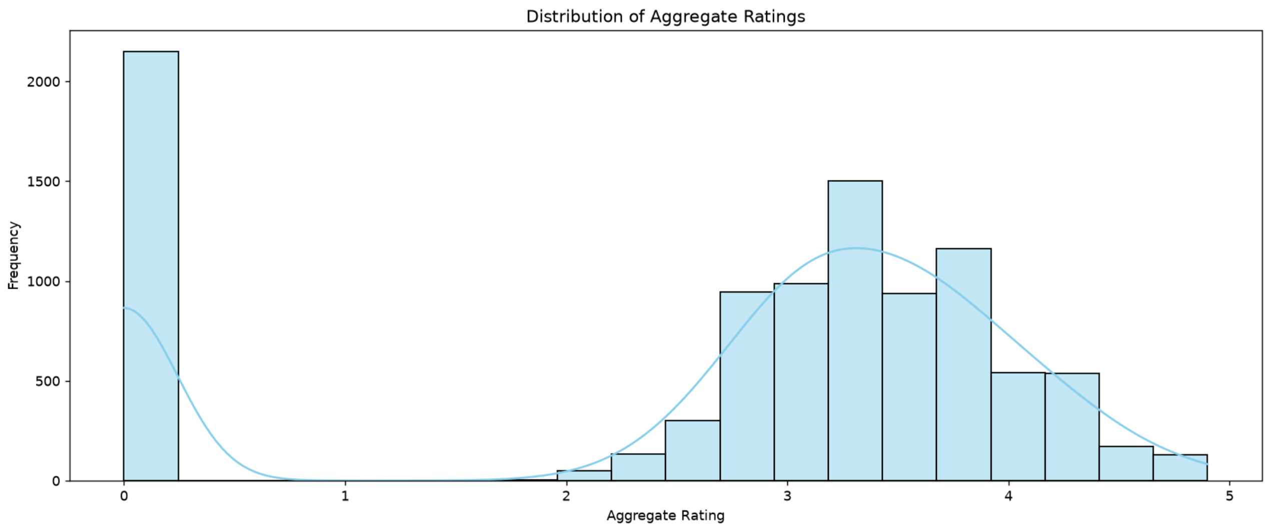
I created five charts using Matplotlib and Seaborn, all with titles and axis labels.

Chart 1: Top 10 cities with the highest number of restaurants



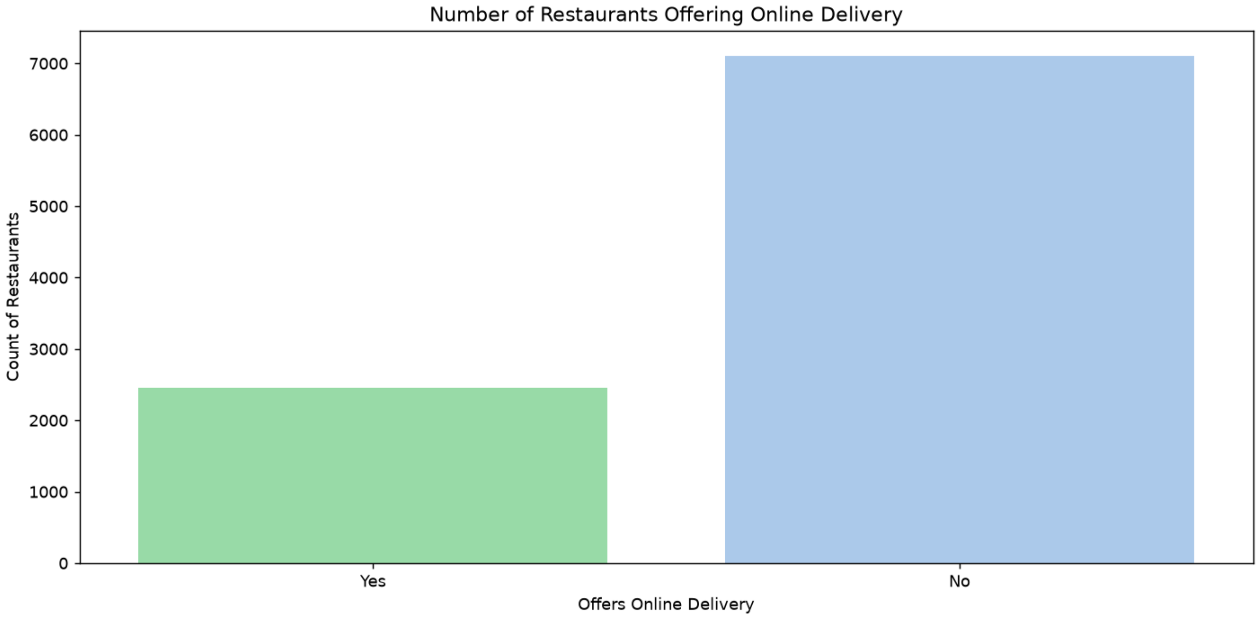
New Delhi stands out with the most restaurants, followed by Gurgaon and Noida. The restaurants are heavily concentrated in the Delhi NCR region and all other cities trail far behind.

Chart 2: Distribution of aggregate ratings



The ratings are bi-modal. Most restaurants are grouped around 3.0 to 3.5, and there is another smaller group around 4.5 and above (these are mostly restaurants that have no user ratings). Ratings near 4.0 are less common.

Chart 3: Number of restaurants offering online delivery

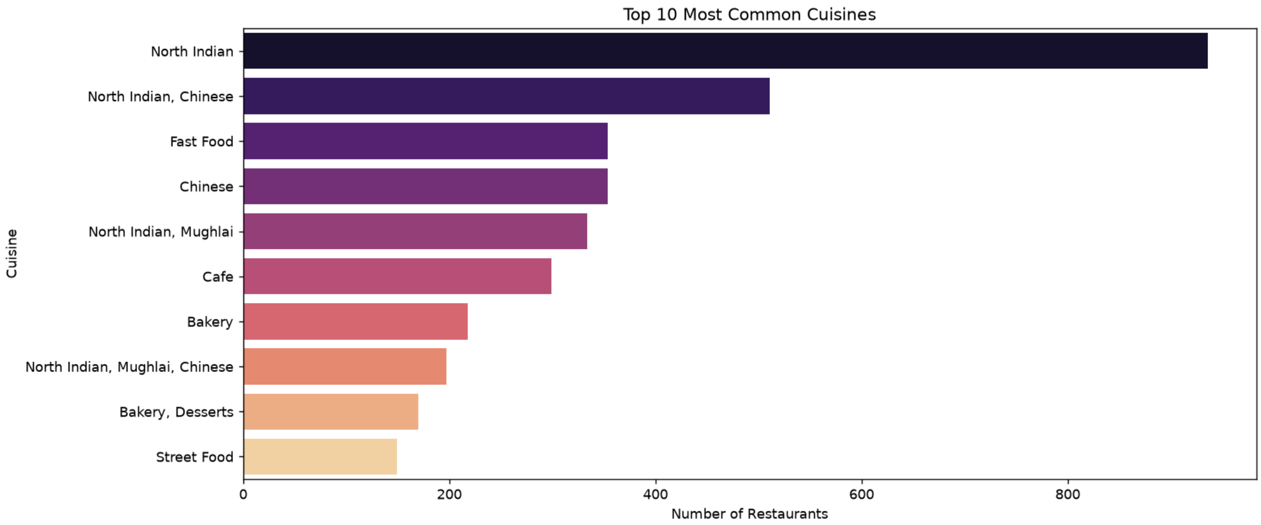


Most restaurants do not offer online delivery. Out of the dataset, 7099 restaurants say No and only 2451 say Yes. So delivery is not the main channel in this data.

Online delivery availability counts

Offers online delivery	Restaurants
No	7099
Yes	2451

Chart 4: Top 10 most common cuisines

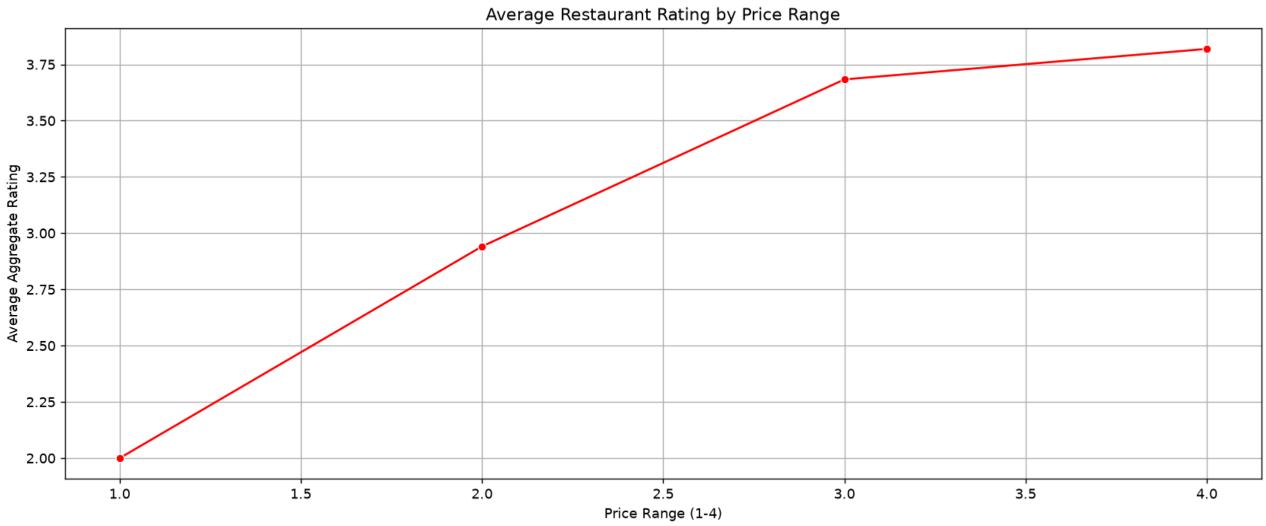


North Indian is the most common cuisine with 936 restaurants, followed by the North Indian + Chinese combination with 511. Fast Food and Chinese come next with 354 each. Multi-cuisine combos are very common, which matches how restaurant menus usually are.

Top 10 cuisines by number of restaurants

Cuisine	Restaurants
North Indian	936
North Indian, Chinese	511
Fast Food	354
Chinese	354
North Indian, Mughlai	334
Cafe	299
Bakery	218
North Indian, Mughlai, Chinese	197
Bakery, Desserts	170
Street Food	149

Chart 5: Average restaurant rating by price range



The average rating increases steadily as the price range goes from 1 to 4. The most expensive restaurants get the highest average ratings, so it looks like pricier restaurants tend to receive better reviews.

Average rating by price range

Price range	Average rating
1	2.00
2	2.94
3	3.68
4	3.82

5. Conclusion

Q. Which city has the highest restaurant presence?

Ans. **New Delhi** has the highest presence with **5473 restaurants**, which makes the Delhi NCR region the biggest market in the dataset.

Q. What is the most popular cuisine?

Ans. **North Indian** is the most popular cuisine.

Q. Do restaurants with Online Delivery generally have higher ratings?

Ans. Yes. Their average rating is **3.25** compared to **2.47** for restaurants without delivery, a difference of about **0.78**.

Q. Do restaurants offering Table Booking receive better ratings?

Ans. Yes. They average **3.44** compared to **2.56** for those without, a difference of about **0.88**, which is even bigger than the delivery difference.

Q. What did I learn from analyzing this dataset?

Ans. I learned the whole EDA workflow from start to finish. I loaded a real dataset, handled a messy column where cost was stored as text, checked for missing values and duplicates, computed summary statistics, and converted each question into a grouped aggregation or a chart. The main business takeaway is that higher price range, table booking and online delivery are all linked to better ratings, so restaurants that invest in these things seem to earn more satisfied customers.